2

a $3x-2y=8$, $5x+3y=7$

Solution

Agumented matrix

$$\begin{bmatrix} 3 & -2 & : & 8 \\ 5 & 3 & : & 7 \end{bmatrix}$$

$R_2 \longrightarrow 3R_2 - 5R_1$

$$\sim \begin{bmatrix} 3 & -2 & : & 8 \\ 0 & 19 & : & -19 \end{bmatrix}$$

$\therefore \; 19y = -19$

$y = -1$

Now

$3x - 2(-1) = 8$

or, $3x + 2 = 8$

or $x = \frac{6}{3}$

$\therefore \; x = 2$

b $2x-y=5, \; x-2y=1$

Soln

Agumented Matrix

$$\begin{bmatrix} 2 & -1 & : & 5 \\ 1 & -2 & : & 1 \end{bmatrix}$$

$R_1 \longleftrightarrow R_2$

$$\sim \begin{bmatrix} 1 & -2 & : & 1 \\ 2 & -1 & : & 5 \end{bmatrix}$$

$R_2 \longrightarrow R_2 - 2R_1$

$$\sim \begin{bmatrix} 1 & -2 & : & 1 \\ 0 & 3 & : & 3 \end{bmatrix}$$

From the matrix

$$3y = 3$$
$$\therefore \; y = 1$$

substituting in $2x-y=5$

$$2x - 1 = 5$$
$$\text{or } x = \frac{6}{2}$$
$$\therefore x = 3$$

c $x-2y-3z=-1$, $2x+y+z=6$, $x+3y-2z=13$
Soln

Agumented matrix

$$\begin{bmatrix} 1 & -2 & -3 & : & -1 \\ 2 & 1 & 1 & : & 6 \\ 1 & 3 & -2 & : & 13 \end{bmatrix}$$

$R_2 \longrightarrow R_2 - 2R_1$, $R_3 \longrightarrow R_3 - R_1$

$$\sim \begin{bmatrix} 1 & -2 & -3 & : & -1 \\ 0 & 5 & 7 & : & 8 \\ 0 & 5 & 1 & : & 14 \end{bmatrix}$$

$R_3 \longrightarrow R_3 - R_2$

$$\sim \begin{bmatrix} 1 & -2 & -3 & : & -1 \\ 0 & 5 & 7 & : & 8 \\ 0 & 0 & -6 & : & 6 \end{bmatrix}$$

According to matrix

$-6z = 6$
$\therefore z = -1$

Substituting z, in eqn from R_2

$0x + 5y + 7\times(-1) = 8$
or, $5y - 7 = 8$
or $y = \frac{15}{5}$
$\therefore y = 3$

Finally, substituting both in eq^n from R_1

$$x - 2(3) - 3(-1) = -1$$

$$x - 6 + 3 = -1$$
$$x = -1 + 3$$
$$\therefore x = 4$$

d. $9y - 5x = 5,\ x + z = 1,\ z + 2y = 2$

rearranging

$$-5x + 9y + 0z = 5,\ x + 0y + z = 1,\ 0x + 2y + z = 2$$

Agumented matrix

$$\sim \left[\begin{array}{ccc:c} -5 & 9 & 0 & 5 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 2 \end{array}\right]$$

$$\cancel{R_3 \leftrightarrow R_2 -}$$

$$\cancel{\begin{bmatrix} -5 & \\ 0 & 2 \\ 1 & 0 \end{bmatrix}}$$

$R_1 \longleftrightarrow R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ -5 & 9 & 0 & 3 \\ 0 & 2 & 1 & 2 \end{array}\right]$$

$R_2 \longrightarrow R_2 + 5R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 9 & 5 & 8 \\ 0 & 2 & 1 & 2 \end{array}\right]$$

$R_3 \longrightarrow 9R_3 - 2R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 9 & 5 & 8 \\ 0 & 0 & -1 & 2 \end{array}\right]$$

From R_3

$-z = 2$

$\therefore z = -2$ ——— (i)

From R_2

$9y + 5z = 8$

or $9y - 10 = 8$ ——— [From i, $z = -2$; $5 \times -2 = -10$]

or $9y = 18$

$\therefore y = 2$

Similarly from R_1

$$x + z = 1$$

or, $x - 2 = 1$

$\therefore x = 3$

e $x + y + z = 6, \; x - y + z = 2, \; 2x + y - z = 1$

Soln

Agumented matrix:

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{array}\right]$$

$R_2 \rightarrow R_2 - R_1, \; R_3 \rightarrow R_3 - 2R_1$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{array}\right]$$

$R_3 \rightarrow 2R_3 - R_2$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & -6 & -18 \end{array}\right]$$

From the matrix

$-6z = -18$	$-2y = -4$
$z = 3$	$y = 2$

Now substituting y and z in eqn from R_1

$x + y + z = 6$

or, $x + 2 + 3 = 6$

$\therefore x = 1$

f) $x - 2y - z = -7,\ 2x + y + z = 0,\ 3x - 5y + 8z = 13$

Soln

Agumented matrix:

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -1 & -7 \\ 2 & 1 & 1 & 0 \\ 3 & -5 & 8 & 13 \end{array}\right] \begin{array}{l} \\ R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -1 & -7 \\ 0 & 5 & 3 & 14 \\ 0 & 1 & 11 & 34 \end{array}\right] R_3 \to 5R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & -1 & -7 \\ 0 & 5 & 3 & 14 \\ 0 & 0 & 52 & 156 \end{array}\right]$$

from R_3,

$52z = 156$

$\therefore z = 3$

Now,

from R_2

$5y + 3(3) = 14$

$5y = 14 - 9$

$\therefore y = 1$

Finally From R_1

$$x - (2)(1) - 1(3) = -7$$

or $x = -7 + 2 + 3$

$\therefore x = -2$ //

Q. $2x + 6y = -11$, $6x + 20y - 6z = -3$, $6y - 18z = -1$

Soln

Agumented Matrix

$$\sim \begin{bmatrix} 2 & 6 & 0 & : -11 \\ 6 & 20 & -6 & : -3 \\ 0 & 6 & -18 & : -1 \end{bmatrix}$$

$R_2 \longrightarrow R_2 - 3R_1$

$$\sim \begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 0 & 2 & -6 & : & 30 \\ 0 & 6 & -18 & : & -1 \end{bmatrix}$$

$R_3 \longrightarrow R_3 - 3R_2$

$$\sim \begin{bmatrix} 2 & 6 & 0 & : & -11 \\ 0 & 2 & -6 & : & 30 \\ 0 & 0 & 0 & : & -91 \end{bmatrix}$$

$\because (0 = -91)$ is given by R_3

this system of linear equation has no solution.

h $2x+y+4z=1,\ x+y+z=1,\ 4x+y+10z=1$

soln

Agumented matrix

$$\sim \left[\begin{array}{ccc:c} 2 & 1 & 4 & 1 \\ 1 & 1 & 1 & 1 \\ 4 & 1 & 10 & 1 \end{array}\right]$$

$R_1 \longleftrightarrow R_2$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 2 & 1 & 4 & 1 \\ 4 & 1 & 10 & 1 \end{array}\right]$$

$R_2 \longrightarrow R_2 - 2R_1,\ R_3 \longrightarrow R_3 - 4R_1$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & -3 & 6 & -3 \end{array}\right]$$

$R_3 \longrightarrow R_3 - 3R_2$

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Since R_3 gives $0:0$, this has infinite solutions, hence let $z=k$

Now from R_2

$$-1y + 2k = -1$$

$$\therefore\ y = 2k+1$$

from R_1

$1x + 1 + 2k + k = 1$

$\therefore x = -3k$

Q3. Test consistency and solve

a $x+y+z=6$, $x-y+z=8$, $5x-9y=-3$

Sol^n

In matrix form:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 5 & -9 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 8 \\ -3 \end{bmatrix}$$

$AX = b$

For consistency $\rho(A) = \rho(A:b)$

now Agumented matrix :

$$\sim \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 8 \\ 5 & -9 & 0 & -3 \end{array}\right]$$

$R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - 5R_1$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & 2 \\ 0 & -14 & -5 & -33 \end{array}\right]$$

$R_3 \rightarrow R_3 - 7R_2$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & 2 \\ 0 & 0 & -5 & -47 \end{array}\right]$$

$\rho(A) = 3$, $\rho(A:b) = 3$

Now
from R_3:

$$-5z = -47$$
$$z = \frac{47}{5}$$

Similarly

$$-2y = 2$$
$$\therefore y = -1$$

$$x + (-1) + \frac{-47}{5} = 6$$

or $5x - 5 + 47 = 30$

or, $5x - 42 = 30$

or, $x = \frac{-12}{5}$

b $x+y+z=-2,\ 2x+8y+5z=5,\ x+2y-z=2$

Sol

Agumented matrix

$$\sim\left[\begin{array}{ccc:c} 1 & 1 & 1 & -2 \\ 2 & 8 & 5 & 5 \\ 1 & 2 & -1 & 2 \end{array}\right]$$

$R_2 \longrightarrow R_2 - 2R_1,\ R_3 \longrightarrow R_3 - R_1$

$$\sim\left[\begin{array}{ccc:c} 1 & 1 & 1 & -2 \\ 0 & 6 & 3 & 9 \\ 0 & 1 & -2 & 4 \end{array}\right]$$

$R_3 \longrightarrow 6R_3 - R_2$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & -2 \\ 0 & 6 & 3 & 9 \\ 0 & 0 & -15 & 15 \end{array}\right]$$

$$\rho(A) = \rho(A:b)$$

Now,

from R_3

$$-15z = 15$$
$$z = -1$$

from R_2

$$6y + 3(-1) = 9$$

or $6y - 3 = 9$

$\therefore y = 2$

From

R_1

$$x + (2) + (-1) = -2$$

or $x + 1 = -2$

$\therefore x = -3$

c $x+2y-3z=9$, $2x-y+2z=-8$, $3x-y-4z=3$

Sol

Agumented matrix

$$\sim \left[\begin{array}{ccc:c} 1 & 2 & -3 & 9 \\ 2 & -1 & 2 & -8 \\ 3 & -1 & -4 & 3 \end{array}\right]$$

$R_2 \rightarrow R_2 - 2R_1$

$R_3 \rightarrow R_3 - 3R_1$

$$\sim \left[\begin{array}{ccc:c} 1 & 2 & -3 & 9 \\ 0 & -5 & 8 & -26 \\ 0 & -7 & 5 & -24 \end{array}\right]$$

$R_3 \rightarrow 5R_3 - 7R_2$

$$\left[\begin{array}{ccc:c} 1 & 2 & -3 & 9 \\ 0 & -5 & 8 & -26 \\ 0 & 0 & -31 & 62 \end{array}\right]$$

$\rho(A) = \rho(A:b) = 3$

From R_3,

$$-31z = 62$$
$$\therefore z = -2$$

From R_2

$$-5y + 8(-2) = -26$$
or $5y + 16 = 26$

or $5y = 10$

$\therefore y = 2$

From R_1

$$x + 2(2) \cancel{+} -3(-2) = 9$$
or $x + 4 + 6 = 9$

or $x = 9 - 10$

$\therefore x = -1$

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c. $x - y = 1$, $x + z = -6$, $x + y - 2z = 3$

Soln

Agumented form

$$\sim \left[\begin{array}{ccc:c} 1 & -1 & 0 & 1 \\ 1 & 0 & 1 & 6 \\ 1 & 1 & -2 & 3 \end{array}\right]$$

$R_2 \longrightarrow R_2 - R_1$, $R_3 \longrightarrow R_3 - R_1$

$$\sim \left[\begin{array}{ccc:c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & -7 \\ 0 & 2 & -2 & 2 \end{array}\right]$$

$R_3 \longrightarrow R_3 - 2R_2$

$$\left[\begin{array}{ccc:c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & -7 \\ 0 & 0 & -4 & 16 \end{array}\right]$$

$\rho(A) = \rho(A:b) = 3$

From R_3

$-4z = 16$

$z = -4$

From R_2

$y + (-4) = -7$

$\therefore y = -3$

from R_1

$$x-(-3)=1$$
or $x=-2$

If the following system has non-trivial solution, Find it.

$$2x_1-5x_2+8x_3=0$$
$$-2x_1-7x_2+x_3=0$$
$$4x_1+2x_2+7x_3=0$$

Soln

Agumented matrix

$$\left[\begin{array}{ccc:c} 2 & -5 & 8 & 0 \\ -2 & -7 & 1 & 0 \\ 4 & 2 & 7 & 0 \end{array}\right]$$

$R_2 \longrightarrow R_2+R_1$, $R_3 \longrightarrow R_3-2R_1$

$$\sim\left[\begin{array}{ccc:c} 2 & -5 & 8 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 12 & -9 & 0 \end{array}\right]$$

$R_3 \longrightarrow R_3+R_2$

$$\sim\left[\begin{array}{ccc:c} 2 & -5 & 8 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\rho(A) <$ no. of variables $(2<3)$

let,

$x_3 = k$

then,

From R_2

$-12x_2 + 9k = 0$

$\therefore x_2 = \frac{3}{4}k$

From R_1

$2x_1 - 5 \times \frac{3}{4}k + 8k = 0$

or $\frac{8x_1 - 15k + 32k}{4} = 0$

or $8x_1 = -17k$

$\therefore x_1 = -\frac{17}{8}k$